

7

Formulation of Parabolic, Hyperbolic, and Elliptic-Eigenvalue Problems

In order to understand this chapter, a detailed understanding of the material and notational conventions of Chapter 2 is required.

7.1. PARABOLIC CASE: HEAT EQUATION

The heat equation is the prototypical “parabolic” partial differential equation of mathematical physics (e.g., see Stakgold [1]). The formulation given here generalizes the formulation of Chapter 2, which was restricted to steady heat conduction, to time-dependent heat conduction processes. We view all the data to be functions of time, denoted by t , as well as space. For example, the volumetric heat source ℓ is written as

$$\ell: \Omega \times]0, T[\rightarrow \mathbb{R} \quad (7.1.1)$$

which means ℓ is a function of $x \in \Omega$ and $t \in]0, T[$, the open time interval of length $T > 0$. Likewise, we write $\ell = \ell(x, t)$. Similarly, the boundary data is also time-dependent:

$$q: \Gamma_q \times]0, T[\rightarrow \mathbb{R} \quad (7.1.2)$$

$$h: \Gamma_h \times]0, T[\rightarrow \mathbb{R} \quad (7.1.3)$$

Note that Γ_q and Γ_h are assumed *not* to vary with time. To render the initial/boundary-value problem well posed, an *initial condition* on temperature must also be specified. We denote the initial temperature by

$$u_0 : \Omega \rightarrow \mathbb{R} \quad (7.1.4)$$

Two other material properties come into play: The *density*, ρ , and *capacity*, c ; both are assumed positive functions of $x \in \Omega$.

The **strong form** of the initial/boundary-value problem may now be stated:

$$(S) \left\{ \begin{array}{ll} \text{Given } \ell, q, h, \text{ and } u_0, \text{ as in (7.1.1) through (7.1.4), find } u : \bar{\Omega} \times [0, T] \rightarrow \mathbb{R} \\ \text{such that} \\ \rho c u_{,t} + q_{i,i} = \ell \quad \text{on } \Omega \times]0, T[\quad (\text{heat equation}) & (7.1.5) \\ u = q \quad \text{on } \Gamma_q \times]0, T[& (7.1.6) \\ -q_i n_i = h \quad \text{on } \Gamma_h \times]0, T[& (7.1.7) \\ u(x, 0) = u_0(x) \quad x \in \Omega & (7.1.8) \end{array} \right.$$

Recall from Chapter 2 that

$$q_i = -\kappa_{ij} u_{,j} \quad (7.1.9)$$

where $\kappa_{ij} = \kappa_{ij}(x)$ denotes the conductivity tensor. The notation $u_{,t}$ denotes the time derivative of u (i.e., $u_{,t} = \partial u / \partial t$).

Let $\mathcal{U}$ denote the usual space of weighting functions satisfying zero-temperature boundary conditions. Note that functions in $\mathcal{U}$ do *not* depend on time in any way. Let $\mathcal{S} = \mathcal{S}_t$ denote the space of trial solutions. Note that $\mathcal{S}$ varies as a function of t due to the temperature boundary condition, (7.1.2),

$$\mathcal{S} = \mathcal{S}_t = \left\{ u(\cdot, t) \mid u(x, t) = q(x, t), \quad x \in \Gamma_q, \quad u(\cdot, t) \in H^1(\Omega) \right\} \quad (7.1.10)$$

The **weak formulation** consists of the following:

$$(W) \left\{ \begin{array}{ll} \text{Given } \ell, q, h, \text{ and } u_0, \text{ find } u(t) \in \mathcal{S}_t, \quad t \in [0, T] \text{ such that for all } w \in \mathcal{U}, \\ \boxed{(w, \rho c \dot{u}) + a(w, u) = (w, \ell) + (w, h)_\Gamma} & (7.1.11) \\ (w, \rho c u(0)) = (w, \rho c u_0) & (7.1.12) \end{array} \right.$$

Remark

In the weak formulation, x is suppressed as an argument of u . Thus $u(0)$ represents the function u of x at time 0 [i.e., $u(0) = u(\cdot, 0)$]. The superposed dot is used to denote time differentiation. These notations, among other things, enable us to remove commas from the weak formulation which might be confused with those normally appearing in the bilinear forms. Equation (7.1.11) is the weak formulation of the heat equation (7.1.5) and heat-conduction boundary condition (7.1.7), and (7.1.12) is the weak form of the initial condition (7.1.8).

Exercise 1. Verify the formal equivalence of (S) and (W) by proceeding along the lines of Sec. 2.3. The procedure is virtually identical to Sec. 2.3 if (7.1.11) is written as

$$a(w, u) = (w, \tilde{\ell}) + (w, h)_\Gamma$$

where

$$\tilde{\ell} = \ell - \rho c \dot{u}$$

To develop the analogous Galerkin formulation, we proceed as in Sec. 2.4. The approximate weighting function space, $\mathcal{U}^h$, is identical to before. Functions $u^h \in \mathcal{S}^h$ are built up in the usual way:

$$u^h = v^h + q^h \quad (7.1.13)$$

where $v^h(t) \in \mathcal{U}^h$ (i.e., for each fixed t , $v^h(\cdot, t)$ is a function in $\mathcal{U}^h$) and q^h enables satisfaction of the temperature boundary condition. Note that $q^h(t) \in \mathcal{S}_t^h$. All the functions in (7.1.13) depend upon both space and time, i.e.,

$$u^h(x, t) = v^h(x, t) + q^h(x, t) \quad (7.1.14)$$

The *Galerkin formulation* is stated as follows:

$$(G) \left\{ \begin{array}{l} \text{Given } \ell, q, h, \text{ and } u_0, \text{ find } u^h = v^h + q^h, u^h(t) \in \mathcal{S}_t^h, \text{ such that for all } w^h \in \mathcal{U}^h, \\ \boxed{(w^h, \rho c \dot{v}^h) + a(w^h, v^h) = (w^h, \ell) + (w^h, h)_\Gamma - (w^h, \rho c \dot{q}^h) - a(w^h, q^h)} \\ \hspace{15em} (7.1.15) \\ (w^h, \rho c v^h(0)) = (w^h, \rho c u_0) - (w^h, \rho c q^h(0)) \hspace{5em} (7.1.16) \end{array} \right.$$

Note that all given quantities appear on the right-hand sides of (7.1.15) and (7.1.16). The Galerkin equation given by (7.1.15) is called a *semidiscrete* equation because time is left continuous.

The matrix equations require representations of v^h and q^h in terms of basis functions. These are given by

$$v^h(x, t) = \sum_{A \in n-n_q} N_A(x) d_A(t) \quad (7.1.17)$$

$$q^h(x, t) = \sum_{A \in n_q} N_A(x) q_A(t) \quad (7.1.18)$$

Note that the entire time-dependence is carried by the nodal values (i.e., the d_A 's and q_A 's). The shape functions are identical to those used previously (see Chapters 2 and 3). In particular, they are *not* time-dependent.

Remark

Upon encountering representations such as (7.1.17) and (7.1.18) for the first time, we may wonder if the approximation is valid only when the exact solution is "separable" in x and t (e.g., see [1] for a discussion of the separation-of-variables

technique). This is not the case. Even nonseparable exact solutions may be approximated arbitrarily closely by representations like (7.1.17) and (7.1.18).

To develop the matrix equations, we substitute (7.1.17) and (7.1.18) into (7.1.15) and (7.1.16) and proceed along the same lines as in Secs. 2.4 and 2.5. The end result is the following *matrix problem*:

Given $F:]0, T[\rightarrow \mathbb{R}^{n_{eq}}$, find $d: [0, T] \rightarrow \mathbb{R}^{n_{eq}}$ such that

$$\mathbf{M}\dot{\mathbf{d}} + \mathbf{K}\mathbf{d} = \mathbf{F} \quad t \in]0, T[\quad (7.1.19)$$

$$\mathbf{d}(0) = \mathbf{d}_0 \quad (7.1.20)$$

where

$$\mathbf{M} = \mathbf{A}^{n_{el}} (m^e) \quad (7.1.21)$$

$$\mathbf{m} = [m_{ab}^e] \quad (7.1.22)$$

$$m_{ab}^e = \int_{\Omega^e} N_a \rho c N_b \, d\Omega \quad (7.1.23)$$

$$\mathbf{K} = \mathbf{A}^{n_{el}} (k^e) \quad (7.1.24)$$

$$\mathbf{k}^e = [k_{ab}^e] \quad (7.1.25)$$

$$k_{ab}^e = \int_{\Omega^e} \mathbf{B}_a^T \mathbf{D} \mathbf{B}_b \, d\Omega \quad (7.1.26)$$

$$\mathbf{F}(t) = \mathbf{F}_{\text{nodal}}(t) + \mathbf{A}^{n_{el}} (f^e(t)) \quad (7.1.27)$$

$$\mathbf{f}^e = \{f_a^e\} \quad (7.1.28)$$

$$f_a^e = \int_{\Omega^e} N_a \ell \, d\Omega + \int_{\Gamma_k^e} N_a h \, d\Gamma - \sum_{b=1}^{n_{en}} (k_{ab}^e q_b^e + m_{ab}^e \dot{q}_b^e) \quad (7.1.29)$$

$$\mathbf{d}_0 = \mathbf{M}^{-1} \mathbf{A}^{n_{el}} (\hat{\mathbf{d}}^e) \quad (7.1.30)$$

$$\hat{\mathbf{d}}^e = \{\hat{d}_a^e\} \quad (7.1.31)$$

$$\hat{d}_a^e = \int_{\Omega^e} N_a \rho c u_0 \, d\Omega - \sum_{b=1}^{n_{en}} m_{ab}^e q_b^e(0) \quad (7.1.32)$$

(M)

Remarks

1. If we compare with the steady formulation of Chapter 2, we see that the only new matrix which appears is M , the *capacity matrix*. That M is symmetric and positive-definite follows directly from its definition.

2. Equation (7.1.19) is a coupled system of ordinary differential equations. Thus to solve the initial-value problem, i.e., (7.1.19) and (7.1.20), we need to introduce algorithms for solving systems of ordinary differential equations. This subject is taken up in subsequent chapters.

3. Observe in (7.1.29) that the element capacity matrices come into play in accounting for the effect of specified boundary temperatures.

4. It is common in practice to simplify the specification of initial conditions. That is, rather than employ (7.1.30) through (7.1.32), which emanate from (7.1.16), we directly specify d_0 such that the given nodal values are interpolated [i.e., $d_{0A} = u_0(x_A)$, $A \in \eta - \eta_0$].

5. The element capacity matrix m^e turns out to be virtually identical to the element "mass" matrix, which will be introduced in the next section. Consequently, we shall postpone more detailed consideration of the structure of m^e until later.

6. Recall that $q_b^e(t) = 0$ if degree of freedom b of element e is *not* specified. The same rule applies to $\dot{q}_b^e$.

Exercise 2. The details of arriving at (7.1.19) through (7.1.32) are straightforward given familiarity with the analogous developments in Chapter 2. If the results are not "obvious," the reader should fill in all details in step-by-step fashion.

Example

The one-dimensional heat equation is, of course, just a special case of the general formulation above. A strong form of the initial/boundary-value problem can be set which is a generalization of the one-dimensional model problem considered in Chapter 1. The equations are

$$(S) \begin{cases} \rho c u_{,t} - (\kappa u_{,x})_{,x} = \ell & \text{on }]0, L[\times]0, T[& (7.1.33) \\ u(L, t) = q(t) & t \in]0, T[& (7.1.34) \\ -\kappa u_{,x}(0, t) = h(t) & t \in]0, T[& (7.1.35) \\ u(x, 0) = u_0(x) & x \in]0, L[& (7.1.36) \end{cases}$$

The element arrays are

$$m_{ab}^e = \int_{\Omega^e} N_a \rho c N_b \, d\Omega \quad (7.1.37)$$

$$k_{ab}^e = \int_{\Omega^e} N_{a,x} \kappa N_{b,x} \, d\Omega \quad (7.1.38)$$

$$f_a^e = \int_{\Omega^e} N_a \ell \, d\Omega + N_a(0) \delta_e, \quad h - \sum_{b=1}^{n_{en}} (k_{ab}^e q_b^e + m_{ab}^e \dot{q}_b^e) \quad (7.1.39)$$

7.2 HYPERBOLIC CASE: ELASTODYNAMICS AND STRUCTURAL DYNAMICS

The developments of this section generalize those of Secs. 2.7 through 2.10. The nature of the generalization is similar in format to that of the previous section and the reader should first become familiar with Sec. 7.1 before considering this section even if he or she is uninterested in heat conduction.

The initial conditions this time involve specification of both displacements and velocities. Thus

$$u_{0i} : \Omega \rightarrow \mathbb{R} \quad (7.2.1)$$

and

$$\dot{u}_{0i} : \Omega \rightarrow \mathbb{R} \quad (7.2.2)$$

are given functions for each i , $1 \leq i \leq n_{sd}$. The superposed-dot notation in (7.2.2) is symbolic rather than operational.

The remaining prescribed data are

$$f_i : \Omega \times]0, T[\rightarrow \mathbb{R} \quad (7.2.3)$$

$$q_i : \Gamma_{q_i} \times]0, T[\rightarrow \mathbb{R} \quad (7.2.4)$$

$$h_i : \Gamma_{h_i} \times]0, T[\rightarrow \mathbb{R} \quad (7.2.5)$$

The *density*, $\rho : \Omega \rightarrow \mathbb{R}$, assumed to be positive, needs also to be specified in the present case.

The *strong form* of the initial/boundary-value problem is

$$(S) \left\{ \begin{array}{ll} \text{Given } f_i, q_i, h_i, u_{0i} \text{ and } \dot{u}_{0i}, \text{ as in (7.2.1) through (7.2.5), find } u_i : \bar{\Omega} \times [0, T] \rightarrow \mathbb{R} \text{ such that} \\ \rho u_{i,t} = \sigma_{ij,j} + f_i & \text{on } \Omega \times]0, T[\quad (\text{equation of motion}) \quad (7.2.6) \\ u_i = q_i & \text{on } \Gamma_{q_i} \times]0, T[\quad (7.2.7) \\ \sigma_{ij} n_j = h_i & \text{on } \Gamma_{h_i} \times]0, T[\quad (7.2.8) \\ u_i(x, 0) = u_{0i}(x) & x \in \Omega \quad (7.2.9) \\ u_{i,t}(x, 0) = \dot{u}_{0i}(x) & x \in \Omega \quad (7.2.10) \end{array} \right.$$

Recall that $\sigma_{ij} = c_{ijkl} u_{(k,l)}$ and $c_{ijkl} = c_{ijkl}(x)$. Note that the second time derivative (i.e., acceleration) appears in (7.2.6). This is the reason that two initial conditions are required.

The corresponding *weak formulation* is:¹

$$\begin{aligned}
 (W) \left\{ \begin{array}{l} \text{Given } \ell, q, h, u_0, \text{ and } \dot{u}_0, \text{ find } u(t) \in \mathcal{S}_t, t \in [0, T], \text{ such that for all } w \in \mathcal{U}, \\ \boxed{(w, \rho \ddot{u}) + a(w, u) = (w, \ell) + (w, h)_\Gamma} \quad (7.2.11) \\ (w, \rho u(0)) = (w, \rho u_0) \quad (7.2.12) \\ (w, \rho \dot{u}(0)) = (w, \rho \dot{u}_0) \quad (7.2.13) \end{array} \right.
 \end{aligned}$$

Exercise 1. Verify the formal equivalence of (S) and (W). The arguments of Chapter 2 may be used virtually unaltered if (7.2.11) is written as

$$a(w, u) = (w, \tilde{\ell}) + (w, h)_\Gamma$$

where

$$\tilde{\ell} = \ell - \rho \ddot{u}$$

The *semidiscrete Galerkin formulation of elastodynamics* is:

$$\begin{aligned}
 (G) \left\{ \begin{array}{l} \text{Given } \ell, q, h, u_0, \text{ and } \dot{u}_0, \text{ find } u^h = v^h + q^h, u^h(t) \in \mathcal{S}_t^h, \text{ such that for all } w^h \in \mathcal{U}^h, \\ \boxed{(w^h, \rho \ddot{v}^h) + a(w^h, v^h) = (w^h, \ell) + (w^h, h)_\Gamma - (w^h, \rho \ddot{q}^h) - a(w^h, q^h)} \quad (7.2.14) \\ (w^h, \rho v^h(0)) = (w^h, \rho u_0) - (w^h, \rho q^h(0)) \quad (7.2.15) \\ (w^h, \rho \dot{v}^h(0)) = (w^h, \rho \dot{u}_0) - (w^h, \rho \dot{q}^h(0)) \quad (7.2.16) \end{array} \right.
 \end{aligned}$$

The representations of v^h and q^h are given by

$$v_i^h(x, t) = \sum_{A \in \eta - \eta_{q_i}} N_A(x) d_{iA}(t) \quad (7.2.17)$$

$$q_i^h(x, t) = \sum_{A \in \eta_{q_i}} N_A(x) q_{iA}(t) \quad (7.2.18)$$

These and the usual arguments lead to the *matrix problem*:

Given $F:]0, T[\rightarrow \mathbb{R}^{n_{eq}}$, find $d:]0, T[\rightarrow \mathbb{R}^{n_{eq}}$ such that

$$M \ddot{d} + K d = F \quad t \in]0, T[\quad (7.2.19)$$

$$d(0) = d_0 \quad (7.2.20)$$

$$\dot{d}(0) = \dot{d}_0$$

(7.2.21)

where

$$M = \mathbf{A}^{n_{el}} (m^e)$$

$$m^e = [m_{pq}^e]$$

(7.2.23)

$$m_{pq}^e = \delta_{ij} \int_{\Omega^e} N_a \rho N_b d\Omega$$

(7.2.24)

(Recall $p = n_{ed}(a - 1) + i$ and $q = n_{ed}(b - 1) + j$)

$$K = \mathbf{A}^{n_{el}} (k^e)$$

(7.2.25)

$$k^e = [k_{pq}^e]$$

(7.2.26)

$$k_{pq}^e = \mathbf{e}_i^T \int_{\Omega^e} \mathbf{B}_a^T D \mathbf{B}_b d\Omega \mathbf{e}_j$$

(7.2.27)

(M)

$$\mathbf{F}(t) = \mathbf{F}_{\text{nodal}}(t) + \mathbf{A}^{n_{el}} (f^e(t))$$

(7.2.28)

$$f^e = \{f_p^e\}$$

(7.2.29)

$$f_p^e = \int_{\Omega} N_a f_i d\Omega + \int_{\Gamma_{k_i}^e} N_a h_i d\Gamma - \sum_{q=1}^{n_{eq}} (k_{pq}^e q_q^e + m_{pq}^e \ddot{q}_q^e) \quad (\text{no sum on } i)$$

(7.2.30)

$$d_0 = M^{-1} \mathbf{A}^{n_{el}} (\hat{d}^e)$$

(7.2.31)

$$\hat{d}^e = \{\hat{d}_p^e\}$$

(7.2.32)

$$\hat{d}_p^e = \int_{\Omega^e} N_a \rho u_{0i} d\Omega - \sum_{q=1}^{n_{eq}} m_{pq}^e \ddot{q}_q^e(0)$$

(7.2.33)

$$\dot{d}_0 = M^{-1} \mathbf{A}^{n_{el}} (\hat{\dot{d}}^e)$$

(7.2.34)

$$\hat{\dot{d}}^e = \{\hat{\dot{d}}_p^e\}$$

(7.2.35)

$$\hat{\dot{d}}_p^e = \int_{\Omega^e} N_a \rho \dot{u}_{0i} d\Omega - \sum_{q=1}^{n_{eq}} m_{pq}^e \dot{\ddot{q}}_q^e(0)$$

(7.2.36)

Remarks

1. The main addition to what we have encountered previously in the elastostatics formulation of Chapter 2 is the **mass matrix**, M . The reader should verify that M is symmetric and positive-definite. Except for the Kronecker delta and different material parameter inside the integrand, the mass and capacity matrix [(7.1.21) through (7.1.23)] are identical. To appreciate the origin of the Kronecker delta, we will sketch part of the calculation that leads to the matrix formulation. If we restrict attention to the i th element, then

$$\begin{aligned} (w^h, \rho \ddot{u}^h)^e &= \int_{\Omega^e} w_i^h \rho \ddot{u}_i^h d\Omega \\ &= \delta_{ij} \int_{\Omega^e} w_i^h \rho \ddot{u}_j^h d\Omega \quad (7.2.37) \\ &= \sum_{A,B} c_{iA} \delta_{ij} \int_{\Omega^e} N_A \rho N_B d\Omega \ddot{d}_{jB} \quad (\text{summation of } i \text{ and } j \text{ implied}) \end{aligned}$$

2. Equation (7.2.19) is a coupled system of second-order ordinary differential equations. Algorithms for solving equations of this type are described in Chapter 9.

3. Note that the element mass is involved in the adjustment of forces due to nonzero boundary accelerations (see (7.2.30)).

4. As mentioned in the previous section, we usually simplify the specification of initial conditions in practice. Nodal interpolates in this case take the form

$$d_{0P} = u_{0i}(x_A) \quad (7.2.38)$$

$$\dot{d}_{0P} = \dot{u}_{0i}(x_A) \quad (7.2.39)$$

where $P = M(i, A)$. Recall LM is the array described in Chapter 2.

Exercise 2. Fill in the omitted details leading to the matrix formulation of elastodynamics.

Viscous Damping

In structural dynamics we often work with systems of the form

$$M\ddot{d} + C\dot{d} + Kd = F$$

(7.2.40)

where C is the **viscous damping matrix**. A particularly convenient form of C is the **Rayleigh damping matrix**

$$C = aM + bK$$

(7.2.41)

parameters. The two constituents of Rayleigh damping are seen to

be mass and stiffness proportional. We would like to enlarge our theoretical framework to include Rayleigh damping. The necessary modifications are as follows: Replace the equation of motion, (7.2.6), by

$$\rho u_{i,tt} + a \rho u_{i,t} = \sigma_{ij,j} + f_i \quad (7.2.42)$$

where the generalized Hooke's law is modified to account for the stiffness proportional effect, namely,

$$\sigma_{ij} = c_{ijkl}(u_{(k,l)} + b \dot{u}_{(k,l)}) \quad (7.2.43)$$

In addition to the appearance of the $C\dot{d}$ -term in (7.2.40), the effect of the viscous damping matrix is also felt in modifying the forces due to prescribed displacement boundary conditions. Specifically,

$$f_p^e = \text{right-hand side of (7.2.30)} - \sum_{q=1}^{n_{eq}} c_{pq}^e \dot{\varphi}_q^e \quad (7.2.44)$$

where

$$c^e = a m^e + b k^e \quad (7.2.45)$$

Everything else remains the same. The parameters a and b may be selected to produce desired damping characteristics.

Example

In one dimension, the above formulation leads to a *wave equation*. This also may be viewed as a generalization of the one-dimensional model problem of Chapter 1. Various interpretations are possible. For example, the axial motion of an elastic rod of length L is governed by the equation

$$\rho u_{,tt} = \sigma_{,x} + f \quad \text{on }]0, L[\times]0, T[\quad (7.2.46)$$

where

$$\sigma = E u_{,x} \quad (7.2.47)$$

and $E = E(x)$ is Young's modulus. Boundary and initial conditions may be specified in analogous fashion to Chapter 1, namely,

$$u(L, t) = \varphi(t) \quad t \in]0, T[\quad (7.2.48)$$

$$-E u_{,x}(0, t) = h(t) \quad t \in]0, T[\quad (7.2.49)$$

$$u(x, 0) = u_0(x) \quad x \in]0, L[\quad (7.2.50)$$

$$\dot{u}(x, 0) = \dot{u}_0(x) \quad x \in]0, L[\quad (7.2.51)$$

The resulting element arrays are virtually identical to the ones encountered in the one-dimensional heat equation example at the end of Sec. 7.1, viz.,

$$m_{ab}^e = \int_{\Omega^e} N_a \rho N_b d\Omega \quad (7.2.52)$$

$$k_{ab}^e = \int_{\Omega^e} N_{a,x} E N_{b,x} d\Omega \quad (7.2.53)$$

$$f_a^e = \int_{\Omega^e} N_a f d\Omega - N_a(0) \delta_{e,1} / t - \sum_{b=1}^{n_{en}} (k_{ab}^e q_b^e + m_{ab}^e \ddot{q}_b^e) \quad (7.2.54)$$

Note that in the present case, element equation and node numbers coincide (i.e., $p = a$ and $q = b$). Compare (7.1.37) through (7.1.39) with (7.2.52) through (7.2.54).

Exercise 3. In previous chapters, the element stiffness matrix, (7.2.53), was evaluated (without E) for typical C^0 shape functions. Assuming ρ is constant, evaluate the element mass matrix, m^e , for the following shape functions:

i. Piecewise linears.

$$\left[\text{Answer: } m^e = \frac{\rho h}{6} \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix} \right]$$

ii. Piecewise quadratics.

$$\left[\text{Answer: } m^e = \frac{\rho h}{30} \begin{bmatrix} 4 & 2 & -1 \\ 2 & 16 & 2 \\ -1 & 2 & 4 \end{bmatrix} \right]$$

Exercise 4. The initial/boundary-value problem for the deflection of an *elastic membrane on a Winkler foundation* is stated as follows:

Given $f : \Omega \times]0, T[\rightarrow \mathbb{R}$, $q : \Gamma_q \times]0, T[\rightarrow \mathbb{R}$, $h : \Gamma_h \times]0, T[\rightarrow \mathbb{R}$, $u_0 : \Omega \rightarrow \mathbb{R}$, and $\dot{u}_0 : \Omega \rightarrow \mathbb{R}$, find $u : \bar{\Omega} \times [0, T] \rightarrow \mathbb{R}$ such that

$$\begin{aligned} u_{,ii} - \alpha u + f &= u_{,n} && \text{on } \Omega \times]0, T[\\ u &= q && x \in \Gamma_q, t \in]0, T[\\ u_{,n} &= h && x \in \Gamma_h, t \in]0, T[\\ u &= u_0 && x \in \Omega, t = 0 \\ u_{,t} &= \dot{u}_0 && x \in \Omega, t = 0 \end{aligned}$$

where $u_{,n} = \partial u / \partial n$ is the derivative in the direction of the outward unit normal vector and $\alpha > 0$.

Set up the following finite element paraphernalia:

- A semidiscrete weak form of the problem in which the h -boundary condition is "natural."
- The corresponding Galerkin form of the problem.
- The matrix ordinary-differential-equation problem.

Solution

- Find $u \in \mathcal{S}$, such that for all $w \in \mathcal{U}$

$$(w, \ddot{u}) + a(w, u) = (w, f) + (w, h)_\Gamma$$

$$(w, u(0) - u_0) = 0$$

$$(w, \dot{u}(0) - \dot{u}_0) = 0$$

where

$$(w, \ddot{u}) = \int_{\Omega} w \ddot{u} \, d\Omega$$

$$a(w, u) = \int_{\Omega} (w_{,i} u_{,i} + \alpha w u) \, d\Omega$$

$$(w, \ell) = \int_{\Omega} w \ell \, d\Omega$$

$$(w, h)_{\Gamma} = \int_{\Gamma_h} w h \, d\Gamma$$

$$\vdots$$

- b. Find $u^h = v^h + q^h \in \mathcal{S}^h$, $v^h \in \mathcal{U}^h$ such that for all $w^h \in \mathcal{U}^h$ (as usual)

$$(w^h, \ddot{u}^h) + a(w^h, u^h) = (w^h, \ell) + (w, h)_{\Gamma}$$

$$(w^h, \ddot{v}^h) + a(w^h, v^h) = (w^h, \ell) + (w^h, h)_{\Gamma} - (w^h, \ddot{q}^h) - a(w^h, q^h)$$

(Likewise for initial conditions.)

- c. $M\ddot{d} + Kd = F$; $d(0) = d_0$; $\dot{d}(0) = \dot{d}_0$

where

$$m_{ab}^e = \int_{\Omega^e} N_a N_b \, d\Omega$$

$$k_{ab}^e = \int_{\Omega^e} (N_{a,i} N_{b,i} + \alpha N_a N_b) \, d\Omega$$

$$f_a^e = \int_{\Omega^e} N_a \ell \, d\Omega + \int_{\Gamma_h^e} N_a h \, d\Gamma - \sum_{b=1}^{n_{en}} (m_{ab}^e \ddot{q}_b^e + k_{ab}^e q_b^e)$$

(Usual assembly algorithm plus initial conditions.)

7.3 EIGENVALUE PROBLEMS: FREQUENCY ANALYSIS AND BUCKLING

To each of the classes of time-dependent problems considered in Sec. 7.1 and 7.2 there are corresponding eigenvalue problems. Eigenvalue problems play particularly important roles in the analysis of elastic solids and structures. The natural frequencies and mode shapes of free vibration are determined by eigenvalue problems as are buckling loads and modes. Some examples illustrating potential applications are as follows.

Free Vibration of an Elastic Rod

The frequencies and mode shapes are governed by the following equation:

$$\lambda \rho u + (Eu_{,x})_{,x} = 0 \quad x \in]0, L[\quad (7.3.1)$$

Various homogeneous boundary conditions may be employed. For example, corresponding to our "standard" one-dimensional case we have

$$u(L) = 0 \quad (\text{fixed}) \quad (7.3.2)$$

$$-Eu_{,x}(0) = 0 \quad (\text{free}) \quad (7.3.3)$$

The nontrivial solutions of (7.3.1) through (7.3.3) are countably infinite. That is, for each $k = 1, 2, \dots, \infty$, there is an eigenvalue λ_k and corresponding eigenfunction $u_{(k)}:]0, L[\rightarrow \mathbb{R}$ which satisfy (7.3.1) through (7.3.3). It is a simple exercise to show that each λ_k is positive, $0 < \lambda_1 \leq \lambda_2 \leq \lambda_3 \leq \dots$, and that the eigenfunctions are orthogonal in the sense that

$$\int_0^L u_{(k)} \rho u_{(l)} dx = \delta_{kl} \quad (7.3.4)$$

The eigenvalue $\lambda_k = \omega_k^2$, where ω_k is the k th natural frequency and $u_{(k)}$ is the corresponding normal mode shape. The eigenfunctions are determined only up to a multiplicative constant. The arbitrariness may be removed by the normalization

$$\int_0^L \rho u^2 dx = 1 \quad \text{or} \quad \max_{x \in [0, L]} |u(x)| = 1.$$

A *weak formulation* of this problem is given by the following:

$$(W) \left\{ \begin{array}{l} \text{Find all eigenpairs } \{\lambda, u\}, u \in \mathcal{S}(\equiv \mathcal{U}), \text{ such that for all } w \in \mathcal{U} \\ \boxed{a(w, u) - \lambda(w, \rho u) = 0} \quad (7.3.5) \\ \text{where} \\ a(w, u) = \int_0^L w_{,x} E u_{,x} dx \quad (7.3.6) \\ (w, \rho u) = \int_0^L w \rho u dx \quad (7.3.7) \end{array} \right.$$

The essential boundary condition, $u(L) = 0$, needs to be built into the definition of $\mathcal{U}$.

The equivalence of strong and weak forms follows from by now standard arguments. Note

$$a(w, u) - \lambda(w, \rho u) = - \int_0^L w (\lambda \rho u + (Eu_{,x})_{,x}) dx - (w E u_{,x}) \Big|_{x=0} \quad (7.3.8)$$

The remaining details are left as an exercise for the reader.

The corresponding **Galerkin formulation** is as follows:

Find all $\lambda^h \in \mathbb{R}$ and $u^h \in \mathcal{S}^h (\equiv \mathcal{U}^h)$ such that for all $w^h \in \mathcal{U}^h$

$$a(w^h, u^h) - \lambda^h(w^h, \rho u^h) = 0 \quad (7.3.9)$$

Substitution of shape-function expansions for w^h and u^h gives rise to the **matrix eigenvalue problem**.

Find eigenvalues λ_k^h and eigenvectors ψ_k , $k = 1, 2, \dots, n_{eq}$, that satisfy

$$(K - \lambda_k^h M)\psi_k = 0 \quad (\text{no sum on } k) \quad (7.3.10)$$

The eigenvectors ψ_k contain the nodal values of $u_{(k)}^h$. That is, the A th component of ψ_k is $u_{(k)}^h(x_A)$. In terms of the matrix problem, the orthogonality condition may be expressed as [see (7.3.4)]

$$\psi_k^T M \psi_l = \delta_{kl} \quad (7.3.11)$$

The element matrices that contribute to M and K are defined by (7.2.52) and (7.2.53), respectively.

Buckling of a Thin Beam ("Elastic Instability")

Consider a thin (i.e., Bernoulli-Euler) beam subjected to simply supported boundary conditions (see Fig. 7.3.1). We assume the beam is subjected to a uniform axial compression, λ , and wish to calculate the critical value of λ (i.e., "buckling load"). The equations are

$$\lambda u_{,xx} + (EI u_{,xx})_{,xx} = 0 \quad (7.3.12)$$

$$u(0) = u(L) = 0 \quad (7.3.13)$$

$$u_{,xx}(0) = u_{,xx}(L) = 0 \quad (7.3.14)$$

There are an infinite number of eigenpairs that satisfy (7.3.12) through (7.3.14). The smallest eigenvalue, $\lambda_1 > 0$, is the critical buckling load and the corresponding eigenfunction, $u_{(1)}$, is the buckling mode. An exact analysis reveals that $\lambda_k = k^2 \pi^2 EI / L^2$ and $u_{(k)} = \sin k\pi x / L$.

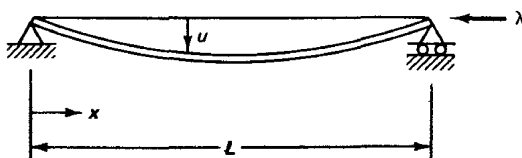


Figure 7.3.1 Simply supported beam under axial compression.

A **weak formulation** of the problem is:

$$\begin{aligned}
 (W) \left\{ \begin{array}{l} \text{Find } u \in \mathcal{S}(\equiv \mathcal{U}) \text{ such that for all } w \in \mathcal{U} \\ \boxed{a(w, u) - \lambda b(w, u) = 0} \quad (7.3.15) \\ \text{where} \\ a(w, u) = \int_0^L w_{,xx} E l u_{,xx} dx \quad (7.3.16) \\ b(w, u) = \int_0^L w_{,x} u_{,x} dx \quad (7.3.17) \end{array} \right.
 \end{aligned}$$

An appropriate space of functions consists of H^2 -functions (see Sec. 1.16) with (7.3.13) as essential boundary conditions. Integration by parts and $w(0) = w(L) = 0$ reveals

$$a(w, u) - \lambda b(w, u) = \int_0^L w(\lambda u_{,xx} + (E l u_{,xx})_{,xx}) dx + (w_{,x} E l u_{,xx}) \Big|_0^L \quad (7.3.18)$$

The equivalence of strong and weak forms may be argued in the usual way with the aid of this expression.

The corresponding **Galerkin formulation** is:

Find $u^h \in \mathcal{S}^h(\equiv \mathcal{U}^h)$ such that for all $w^h \in \mathcal{U}^h$,

$$\boxed{a(w^h, u^h) - \lambda^h b(w^h, u^h) = 0} \quad (7.3.19)$$

The C^1 -continuous Hermite cubics (see Sec. 1.16) serve to define $\mathcal{U}^h$.

The matrix problem has the same form as (7.3.10). The element contributions are defined by

$$\boxed{k_{pq}^e = \int_{\Omega^e} N_{p,xx} E l N_{q,xx} dx} \quad (7.3.20)$$

$$\boxed{m_{pq}^e = \int_{\Omega^e} N_{p,x} N_{q,x} dx} \quad (7.3.21)$$

Strictly speaking, (7.3.21) is not a mass matrix. It is usually referred to as an **initial-stress**, or **geometric, stiffness** in this context.

Exercise 1. Evaluate m^e [i.e., (7.3.21)] using Hermite cubics. (Observe that three-point Gauss quadrature enables exact integration.) Recall that the exact expression for k^e was obtained in Sec. 1.16.

Free Vibration of a Thin Beam

The vibration of a thin beam is governed by

$$-\lambda \rho A u + (EI u_{,xx})_{,xx} = 0 \quad (7.3.22)$$

where $A = A(x)$ is the cross-sectional area. Comparison with (7.3.12) reveals that in place of $\lambda u_{,xx}$, the term $-\lambda \rho A u$ appears. In the simply supported case, the weak formulation is

$$a(w, u) - \lambda(w, \rho A u) = 0 \quad (7.3.23)$$

where $a(w, u)$ is again defined by (7.3.16) and

$$(w, \rho A u) = \int_0^L w \rho A u \, dx \quad (7.3.24)$$

The matrix problem takes the form (7.3.10) and the element arrays are defined by (7.3.20) and

$$m_{pq}^e = \int_{\Omega^e} N_p \rho A N_q \, dx \quad (7.3.25)$$

Exercise 2. Evaluate (7.3.25) using Hermite cubics.

7.3.1 Standard Error Estimates

For the types of problems considered in Chapters 1 through 3, standard error estimates are available for eigenvalues and eigenfunctions. These estimates are the counterparts of those given in Sec. 4.1. The more-sophisticated methods presented after Sec. 4.1 and in Chapters 5 and 6 lie outside the realm of these estimates and therefore must be considered on a separate basis. The present estimates indicate the optimality of the Galerkin finite element method for the standard classes of elliptic eigenvalue problems. The main results are [2]:

$$\lambda_l \leq \lambda_l^h \leq \lambda_l + ch^{2(k+1-m)} \lambda_l^{(k+1)/m} \quad (7.3.26)$$

$$\|u_{(l)}^h - u_{(l)}\|_m \leq ch^{(k+1-m)} \lambda_l^{(k+1)/2m} \quad (7.3.27)$$

where c is a constant independent of h and λ_l .

Remarks

1. Note that the first inequality in (7.3.26) says that the l th approximate eigenvalue is bounded from below by the l th exact eigenvalue. This “upper-bound” property is *not* preserved once the Galerkin rules are violated (e.g., when reduced integration or incompatible modes are employed).

2. The rate of convergence (i.e., power of h) of eigenvalues is *twice* that of eigenfunctions in the H^m -norm [compare (7.3.26) with (7.3.27)].

3. The appearance of powers of the eigenvalue on the right-hand sides of (7.3.26) and (7.3.27) suggests that the quality of approximation *deteriorates* for higher modes (recall $0 < \lambda_1 \leq \lambda_2 \leq \lambda_3 \leq \dots$). This will be shown to be the case in specific examples.

4. In frequency analysis we are interested in the error in $\omega_l^h = (\lambda_l^h)^{1/2}$. It turns out that the order of error in the frequency is the same as that in the eigenvalue. This can be seen as follows: Let

$$\epsilon = \frac{\lambda_l^h}{\lambda_l} - 1 \quad (7.3.28)$$

Then,

$$\frac{\omega_l^h}{\omega_l} = \left(\frac{\lambda_l^h}{\lambda_l} \right)^{1/2} = (1 + \epsilon)^{1/2} = 1 + \frac{\epsilon}{2} + O(\epsilon^2) \quad (7.3.29)$$

and so

$$\frac{\omega_l^h}{\omega_l} - 1 = O(\epsilon) = O(h^{2(k+1-m)}) \quad (7.3.30)$$

Thus we can expect the same rate of convergence for frequencies and eigenvalues.

5. An L_2 -estimate for eigenfunctions is also available [2]:

$$\|u_{(l)}^h - u_{(l)}\|_0 \leq ch^\sigma \lambda_l^{(k+1)/(2m)} \quad (7.3.31)$$

where

$$\sigma = \min\{k + 1, 2(k + 1 - m)\} \quad (7.3.32)$$

6. The accuracy of the eigenvalues and eigenfunctions are measures of the quality of both M and K , in other words, the entire spatial discretization.

Example 1

Consider an elastic boundary value problem ($m = 1$) and assume linear elements are employed ($k = 1$). The error estimates take the form:

$$\frac{\omega_l^h}{\omega_l} - 1 = O(h^2)$$

$$\|u_{(l)}^h - u_{(l)}\|_1 = O(h)$$

For quadratic-level elements ($k = 2$) we have

$$\frac{\omega_l^h}{\omega_l} - 1 = O(h^4)$$

$$\|u_{(l)}^h - u_{(l)}\|_1 = O(h^2)$$

Example 2

Consider the Hermite cubic beam element ($m = 2, k = 3$):

$$\frac{\omega_l^h}{\omega_l} - 1 = O(h^4)$$

$$\|u_{(l)}^h - u_{(l)}\|_2 = O(h^2)$$

Remark

If numerical quadrature is employed to integrate the element mass and/or stiffness, the same conditions apply as described in Sec. 4.1. That is, a rule accurate enough to exactly integrate all monomials through degree $\bar{k} + k - 2m$ is sufficient to maintain full rate of convergence. (Recall $\bar{k}$ is the order of the highest-order monomial appearing in the element shape functions.) A sufficient condition for convergence is that the rule exactly integrate monomials through degree $\bar{k} - m$.

Exercise 3. Use the “minimax” characterization of eigenvalues, i.e.,

$$\lambda_l = \min_{E_l} \left\{ \max_{u \in E_l} \mathcal{R}(u) \right\} \quad (7.3.33)$$

$$\lambda_l^h = \min_{E_l^h} \left\{ \max_{u^h \in E_l^h} \mathcal{R}(u^h) \right\} \quad (7.3.34)$$

where $\mathcal{R}$ denotes the *Rayleigh quotient*, defined by

$$\mathcal{R}(u) = \frac{a(u, u)}{(u, \rho u)} \quad (7.3.35)$$

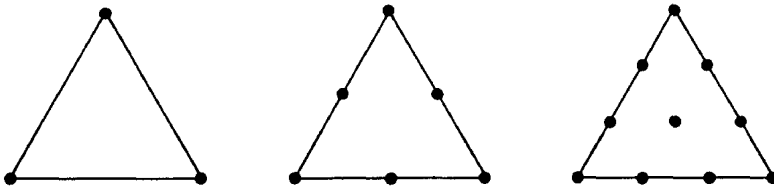
and E_l and E_l^h are l -dimensional subspaces of H^m and $\mathcal{S}^h$, respectively, to establish $\lambda_l \leq \lambda_l^h, l = 1, 2, \dots, n_{eq}$. (The bilinear form in the denominator of (7.3.35) may be $b(u, u)$, $(u, \rho u)$, etc., as in previously discussed cases.) This result is obtained in [2].

Exercise 4. An isoparametric finite element is used in a series of convergence rate studies on linear elastic eigenvalue problems. All tests indicate that the fundamental frequency behaves as follows:

$$\log\left(\frac{\omega^h}{\omega} - 1\right) \sim 2 \log h$$

where ω is the exact frequency, ω^h is the finite element frequency, and h is the mesh parameter.

a. Which of the following elements was used in the study?



Justify your answer.

b. What is the rate of convergence of eigenvectors in the L_2 and H^1 norms?

Exercise 5. Consider the application of bilinear quadrilateral elements to frequency calculations in elasticity. The mass matrix is to be constructed using a Gaussian quadrature formula. The choice is between the one-point, 2×2 , and 3×3 rules. Pick the rule involving the minimum number of points that guarantees that the full rate of convergence of consistent mass is maintained. Explain how you arrived at your answer.

Solution for Exercise 5 The rule must exactly integrate polynomials of degree $\bar{k} + k - 2m = 2 + 1 - 2 = 1$ (i.e., constants and linear terms must be integrated exactly). The *one-point Gauss rule* achieves this accuracy.

7.3.2 Alternative Definitions of the Mass Matrix; Lumped and Higher-order Mass

The recipe given by the Galerkin formulation for the mass matrix is sometimes referred to as variationally “consistent.” Prior to establishment of the variational foundations of the finite element method, mass matrices were concocted in ad hoc fashion. Archer [3] is generally acknowledged as having first pointed out the correctness of *consistent mass*. The consistent-mass matrix leads to the optimal error estimates described in the preceding section. However, as is often the case in finite element analysis, there is strong motivation for breaking the basic rules. This is particularly true in the development of mass matrices. In practice, diagonal or “lumped” mass matrices are often employed due to their general economy and because they lead to some especially attractive time-integration schemes (so-called explicit methods; see Chapters 8 and 9). There are several ways of going about the construction of lumped-mass matrices. One way is to employ nodal quadrature rules.

Mass Lumping by Nodal Quadrature

An integration formula with integration points at the nodes takes the form (see Chapter 3):

$$\int_{\square} g(\xi) d\xi \equiv \sum_{a=1}^{n_{eq}} g(\xi_a) W_a \quad (7.3.36)$$

The effect of such a rule on the mass matrix is to diagonalize it, viz.,

$$\begin{aligned}
 m_{pq}^e &= \delta_{ij} \int_{\Omega^e} N_a \rho N_b d\Omega \\
 &= \delta_{ij} \int_{\square} N_a \rho N_{bj} d\xi \\
 &\equiv \delta_{ij} \sum_{c=1}^{n_{eq}} \underbrace{N_a(\xi_c)}_{\delta_{ac}} \rho(\xi_c) \underbrace{N_b(\xi_c) j(\xi_c)}_{\delta_{bc}} W_c \\
 &= \begin{cases} \delta_{ij} \rho(\xi_a) j(\xi_a) W_a & a = b \\ 0 & a \neq b \end{cases} \quad (\text{no sum on } a)
 \end{aligned} \tag{7.3.37}$$

A theory of numerical quadrature presented in [2] and described earlier defines under what conditions the nodal quadrature mass matrix is convergent and retains full order of accuracy. In the one-dimensional case, full rate of convergence is maintained if the rule is capable of exactly integrating polynomials of degree $2(k - m)$; the minimum condition for convergence is a rule that can exactly integrate polynomials of degree $k - m$. (It should be pointed out that these are *sufficient* conditions. They may not be *necessary*.)

Example 1

Consider piecewise linear shape functions in one dimension and assume $\rho = \text{constant}$. The trapezoidal rule, namely,

$$\xi_1 = -1, \quad \xi_2 = 1, \quad W_1 = W_2 = 1$$

is sufficiently accurate to maintain full rate of convergence (i.e., $2(k - m) = 2(1 - 1) = 0$; thus only constants need to be exactly integrated). It may be verified that in this case

$$m_{\text{lumped}}^e = \frac{\rho h^e}{2} \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \tag{7.3.38}$$

Example 2

Consider piecewise quadratics. Simpson's rule,

$$\xi_1 = -1, \quad \xi_2 = 0, \quad \xi_3 = 1, \quad W_1 = W_3 = \frac{1}{3}, \quad W_2 = \frac{4}{3}$$

is capable of exactly integrating cubic polynomials and so attains full rate of convergence. (The requirement is $2(k - m) = 2(2 - 1) = 2$; i.e., quadratic polynomials.) The lumped-mass matrix is (verify!):

$$m_{\text{lumped}}^e = \frac{\rho h^e}{6} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 4 & 0 \\ 0 & 0 & 1 \end{bmatrix} \tag{7.3.39}$$

Numerical confirmation of the convergence rate of the fundamental frequency is presented in Fig. 7.3.2. Frequency spectra are presented in Fig. 7.3.3. The upper-bound property of consistent mass (i.e., $\omega^h/\omega \geq 1$) is clearly in evidence in Fig. 7.3.3.

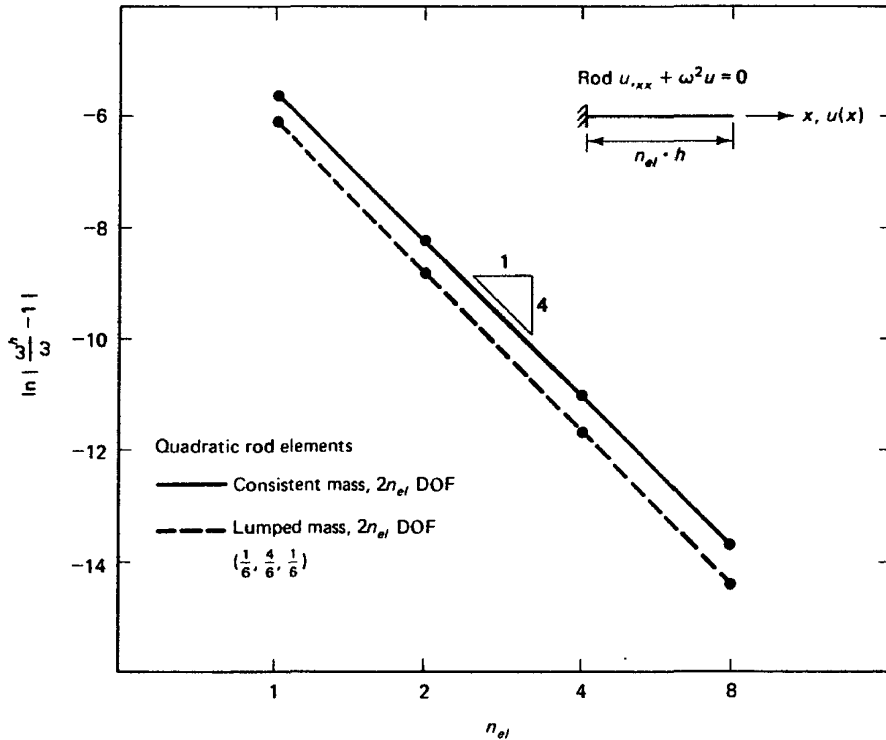


Figure 7.3.2 Convergence of the fundamental frequency for quadratic rod elements (Hughes et al. [4]).

Example 3

For the standard four-node cubic element we need a rule capable of exactly integrating quartic polynomials (i.e., $2(k - m) = 2(3 - 1) = 4$). Unfortunately, for the usual setup in ξ -space, this is impossible. The restriction of interior nodal placement at $\xi = \pm \frac{1}{3}$ limits the order of convergence to cubic:

g	$\int_{-1}^{+1} g \, d\xi$	$\sum_{a=1}^4 g(\xi_a) W_a$	
1	2	$= W_1 + W_2 + W_3 + W_4$	$\Rightarrow \begin{cases} W_1 = \frac{1}{4} \\ W_2 = \frac{3}{4} \\ W_3 = W_2 \\ W_4 = W_1 \end{cases}$
ξ	0	$= -W_1 - \frac{1}{3}W_2 + \frac{1}{3}W_3 + W_4$	
ξ^2	$\frac{2}{3}$	$= W_1 + \frac{1}{9}W_2 + \frac{1}{9}W_3 + W_4$	
ξ^3	0	$= -W_1 - \frac{1}{27}W_2 + \frac{1}{27}W_3 + W_4$	
ξ^4	$\frac{2}{5}$	$\neq W_1 + \frac{1}{81}W_2 + \frac{1}{81}W_3 + W_4$	

Example 4

Let us reconsider the four-node cubic element. We need one additional parameter in order to create a nodal rule that exactly integrates quartic polynomials. If we relax the condition that the interior nodes are located at $\xi = \pm \frac{1}{3}$, a sufficiently accurate rule can

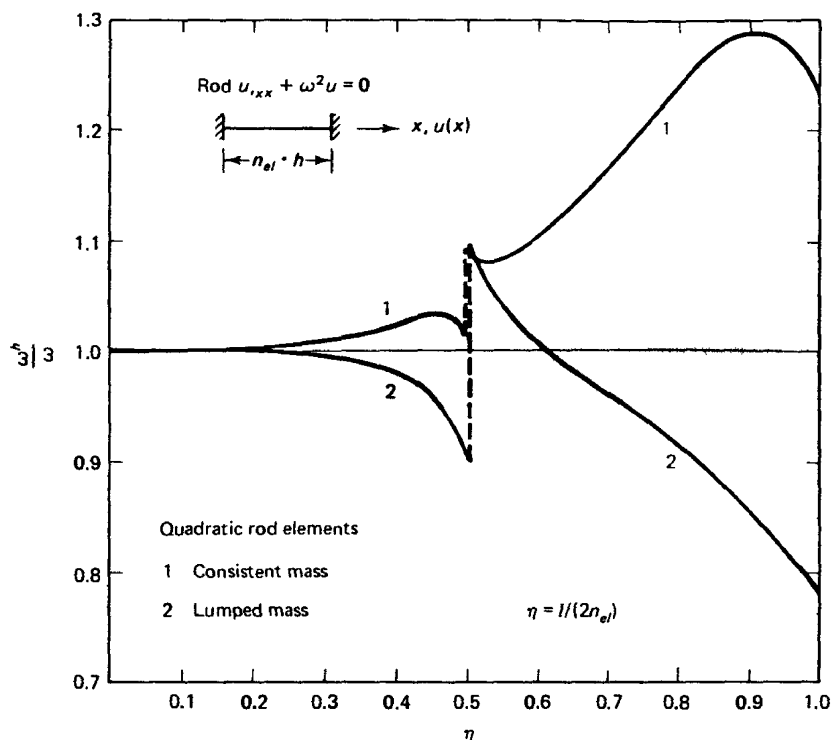


Figure 7.3.3 Frequency spectra for quadratic rod elements (Hughes et al. [4]). η is the mode number.

be developed. Consider the ξ -space setup shown in Fig. 7.3.4. The object is to pick c along with the weights so that an arbitrary quartic polynomial is exactly integrated. The idea is similar to that which led to the Gauss rules. Assume $W_2 = W_1$ and $W_3 = W_2$. Then

g	$\int_{-1}^{+1} g \, d\xi$	$W_1(g(-1) + g(+1)) + W_2(g(-c) + g(+c))$
1	2	$= 2(W_1 + W_2)$
ξ	0	$= 0$
ξ^2	$\frac{2}{3}$	$= 2(W_1 + c^2 W_2)$
ξ^3	0	$= 0$
ξ^4	$\frac{2}{5}$	$= 2(W_1 + c^4 W_2)$
ξ^5	0	$= 0$

There are three equations for the three unknowns. The first equation is used to eliminate

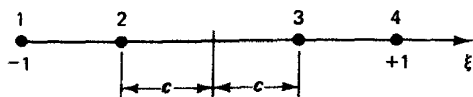


Figure 7.3.4

W_1 from the second and third. Dividing the third equation by the second yields $c^2 = \frac{1}{5}$. Thus $W_1 = \frac{1}{6}$ and $W_2 = \frac{5}{6}$. The mass then is

$$m_{\text{lumped}}^e = \frac{\rho h^e}{12} \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 5 & 0 & 0 \\ 0 & 0 & 5 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \quad (7.3.40)$$

This element mass was originally developed by Fried and Malkus [5]. Their numerical tests confirmed the theoretically predicted convergence rate (i.e., $\lambda^h - \lambda = O(h^6)$).

Note that the relocation of the interior nodes necessitates the derivation of new shape functions.

Remark

The rule derived above is an example of a *Lobatto quadrature rule*. The Lobatto rules are similar to the Gauss rules, except in Lobatto rules the end points of the interval are always included. The locations of the interior points are determined to maximize the accuracy of the rule. The trapezoidal rule and Simpson's rule are the first two Lobatto rules. Lobatto rules for $n_{en} = 3$ through $n_{en} = 10$ are listed in Table 7.3.1.

TABLE 7.3.1 Lobatto quadrature rules

	$\pm \xi_a$	W_a		$\pm \xi_a$	W_a
3	1.00000 000	0.33333 333	7	1.00000 000	0.04761 904
	0.00000 000	1.33333 333		0.83022 390	0.27682 604
				0.46884 879	0.43174 538
				0.00000 000	0.48761 904
4	1.00000 000	0.16666 667	8	1.00000 000	0.03571 428
	0.44721 360	0.83333 333		0.87174 015	0.21070 422
				0.59170 018	0.34112 270
				0.20929 922	0.41245 880
5	1.00000 000	0.10000 000	9	1.00000 00000	0.02777 77778
	0.65465 367	0.54444 444		0.89975 79954	0.16549 53616
	0.00000 000	0.71111 111		0.67718 62795	0.27453 87126
				0.36311 74638	0.34642 85110
				0.00000 00000	0.37151 92744
6	1.00000 000	0.06666 667	10	1.00000 00000	0.02222 22222
	0.76505 532	0.37847 495		0.91953 39082	0.13330 59908
	0.28523 152	0.55485 838		0.73877 38651	0.22488 93420
				0.47792 49498	0.29204 26836
				0.16527 89577	0.32753 97612

Exercise 6. Derive the cubic shape functions of the four-node *Lobatto element*.

Example 5

Consider the Hermite cubic beam element. Because the nodes are located at the end points, the trapezoidal rule will produce a lumped mass:

$$m_{\text{lumped}}^e = \frac{\rho h^e}{2} \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} \quad (7.3.41)$$

Note that half the mass is lumped at each translational degree of freedom and that there are zero masses at the rotational (i.e., slope) degrees of freedom. In order for the full rate of convergence of consistent mass to be guaranteed, the rule should be able to integrate exactly polynomials of degree $2(k - m) = 2(3 - 2) = 2$. The trapezoidal rule is capable of integrating only linear polynomials. Nevertheless, computational results suggest full accuracy is achieved, i.e., (see Fig. 7.3.5),

$$\frac{\omega_l^h}{\omega_l} - 1 = O(h^4) \quad (7.3.42)$$

(This does not contradict the numerical integration theory, which provides sufficient, rather than necessary, conditions.)

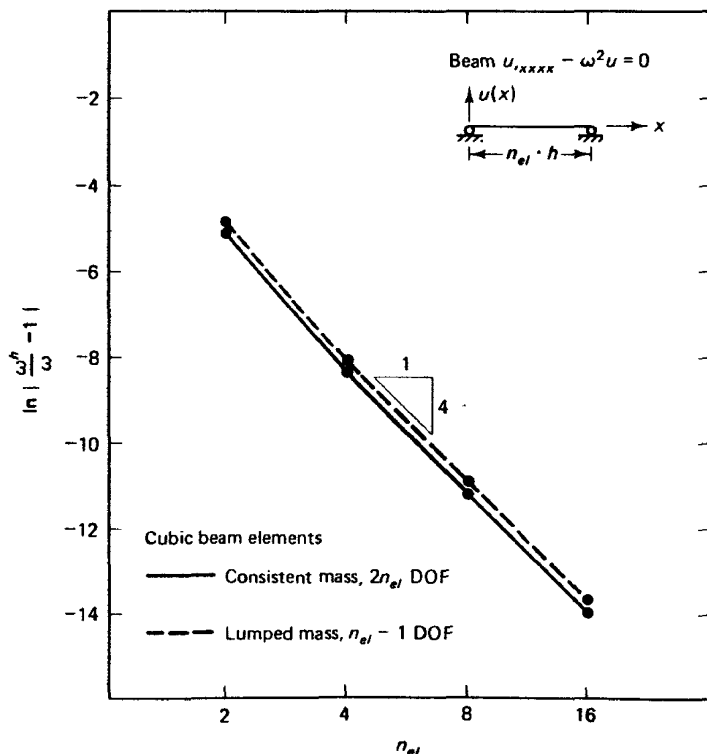


Figure 7.3.5 Convergence of the fundamental frequency for a simply supported beam (Hughes et al. [4]).

Example 6

Optimal quadrature rules have been developed for triangles by Fred and Malkus [5]. The rules are summarized in Fig. 7.3.6.

The theoretical eigenvalue convergence rate for each of the three triangles has been verified numerically in [5]. Note that for the quadratic triangle there are zero masses at the vertices. For the cubic triangle the edge nodes need to be relocated as shown. The vertex masses in this case are negative. Zero and negative masses pose significant impediments in practical applications. (See Remark 1, following the examples.)

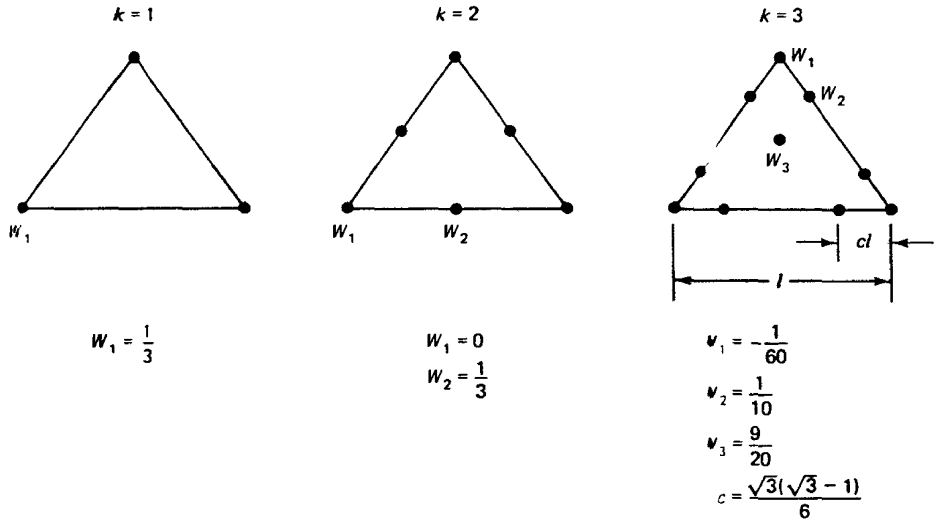
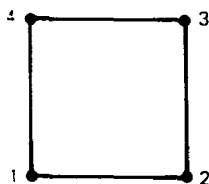


Figure 7.3.6 Optimally accurate nodal quadrature rules for triangles.

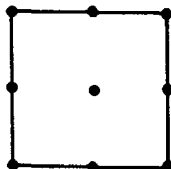
Example 7

For nodal patterns that are products of one-dimensional Lobatto patterns, products of Lobatto rules may be employed. Some examples are illustrated in Fig. 7.3.7 below and on the top of p. 443.



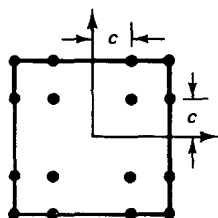
Product trapezoidal

$w_i = 1$, $a = 1, 2, 3, 4$



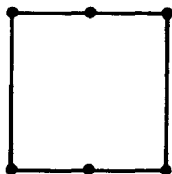
Product Simpson

$w_i = \frac{1}{9}$, corners
 $= \frac{4}{9}$, sides
 $= \frac{16}{9}$, center



Product Lobatto

$$\begin{aligned} W_a &= \frac{1}{36}, \text{ corners} \\ &= \frac{5}{36}, \text{ sides} \\ &= \frac{25}{36}, \text{ interior} \end{aligned}$$


 Trapezoidal $\uparrow \times$ Simpson $\rightarrow$

$$\begin{aligned} W_a &= \frac{1}{3}, \text{ corners} \\ &= \frac{4}{3}, \text{ sides} \end{aligned}$$

Figure 7.3.7 Optimal nodal quadrature rules for product Lobatto patterns.

Example 8

For nodal patterns that are not products of one-dimensional patterns, optimal nodal quadrature rules must be worked out on a case-by-case basis. As an example, consider the eight-node serendipity element (see Fig. 7.3.8). Let the weights assigned to corner and side nodes be denoted W_1 and W_2 , respectively. A rule can be developed that is accurate for all cubic polynomials. This turns out to be of high enough precision to guarantee full rate of convergence (i.e., $\bar{k} + k - 2m = 3 + 2 - 2 = 3$). However, the corner masses are negative and thus this mass matrix is unsuitable for practical use. The calculations leading to the rule are summarized below:

g	$\int_{\square} g \, d\square$	$W_1 \sum_{a=1}^4 g_a + W_2 \sum_{a=5}^8 g_a$
1	4	$= 4(W_1 + W_2)$
ξ	0	$= 0$
η	0	$= 0$
$\xi\eta$	0	$= 0$
ξ^2	$\frac{4}{3}$	$= 4W_1 + 2W_2$
η^2	$\frac{4}{3}$	$= 4W_1 + 2W_2$
ξ^3	0	$= 0$
$\xi^2\eta$	0	$= 0$
$\xi\eta^2$	0	$= 0$
η^3	0	$= 0$

$$\therefore W_1 = -\frac{1}{3}, W_2 = \frac{4}{3}$$

Remarks

1. We have asserted that negative masses are unacceptable in practice. The reason for this can be seen by considering single degree-of-freedom model equations of parabolic and hyperbolic type:

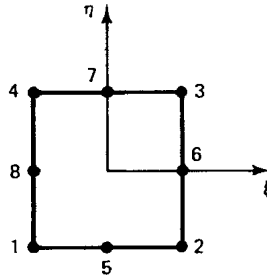


Figure 7.3.8 Eight-node serendipity element.

$$\begin{cases} \dot{d} + \lambda d = 0, & d = \exp(-\lambda t) d_0 & (\text{exponential decay}) \\ -\dot{d} + \lambda d = 0, & d = \exp(\lambda t) d_0 & (\text{unbounded}) \\ \ddot{d} + \omega^2 d = 0, & d = A \sin \omega t + B \cos \omega t, & (\text{oscillatory and bounded}) \\ -\ddot{d} + \omega^2 d = 0, & d = A \sinh \omega t + B \cosh \omega t & (\text{unbounded}) \end{cases}$$

In each case negative mass produces nonphysical behavior. Modes in a multidegree of freedom system will behave in this fashion, vitiating the true solution, when negative masses are present

2. If nodal quadrature is used in axisymmetric analysis, zero masses result along the axis of symmetry for all elements. This can be seen from the following:

$$m_{pq}^e = 2\tau \delta_{ij} \int_{\Omega^e} \rho N_a N_b r \, dr \, dz \quad (7.3.43)$$

$$\equiv \begin{cases} 2\pi \delta_{ij} \rho(\xi_a) r(\xi_a) j(\xi_a) W_a, & a = b \\ 0 & a \neq b \end{cases} \quad (7.3.44)$$

Along the axis of symmetry, $r = 0$, and consequently so are the masses of nodes located there.

As we have seen, due to a tendency to generate zero and negative masses, nodal quadrature is not always an effective tool for producing useful lumped-mass matrices. Thus, other methods have been considered.

Row-sum technique

In the row-sum technique the lumped mass is defined by

$$m'_{ii} = \begin{cases} \delta_{ij} \int_{\Omega^e} \rho N_a \, d\Omega & a = b \\ 0 & a \neq b \end{cases} \quad (7.3.45)$$

The name derives from the fact that

$$\sum_{b=1}^{n_{en}} \int_{\Omega^e} N_a \rho N_b \, d\Omega = \int_{\Omega^e} N_a \rho \underbrace{\left(\sum_{b=1}^{n_{en}} N_b \right)}_1 \, d\Omega = \int_{\Omega^e} N_a \rho \, d\Omega \quad (7.3.46)$$

That is, the elements in each row are summed and lumped on the diagonal.

The row-sum technique eliminates the problem of zero masses at nodes along the z -axis in problems of axisymmetry. However, like nodal quadrature, it sometimes produces negative masses. This is the case for corner nodes of the eight-node serendipity element.

“Special Lumping Technique”

The special lumping technique was developed by Hinton et al. [6]. The name is not too revealing, but the method has an important attribute: *It always produces positive lumped masses*. The idea is to set the entries of the lumped-mass matrix proportional to the diagonal entries of the consistent mass. (By virtue of positive-definiteness, the diagonal entries of consistent mass are necessarily positive.) The constant of proportionality is selected to conserve the total element mass. The formula is

$$m_{pq}^e = \begin{cases} \alpha \delta_{ij} \int_{\Omega^e} \rho N_a^2 d\Omega & a = b \\ 0 & a \neq b \end{cases} \quad (7.3.47)$$

where

$$\alpha = \underbrace{\int_{\Omega^e} \rho d\Omega}_{\text{total element mass}} \bigg/ \underbrace{\left(\sum_{a=1}^{n_{\text{en}}} \int_{\Omega^e} \rho N_a^2 d\Omega \right)}_{\text{sum of diagonal entries of consistent mass}} \quad (7.3.48)$$

The special lumping procedure has been shown numerically to work well on structural and solid mechanics problems [6]. Optimal rates of convergence are typically achieved. Presently, it is the only lumping method that can be recommended for arbitrary elements. Unfortunately, no mathematical theory in support of it has been forthcoming.

Exercise 7. For simple elements, the lumping procedures described above tend to produce similar if not identical masses. Verify this assertion by calculating element masses for the linear triangle and bilinear quadrilateral. Assume the quadrilateral is in the rectangular configuration and in all cases take $\rho = \text{constant}$. Use nodal quadrature, row-sum, and the special lumping techniques.

Exercise 8. Calculate diagonal mass matrices for the one-dimensional three-node element by the row-sum and special lumping techniques: Compare the results with the diagonal mass matrix obtained by Simpson’s rule. Comment.

Remark

Although lumped masses have often been used successfully in solid and struc-

tural mechanics and heat conduction, some disappointing results have been obtained in fluid mechanics (see Gresho et al. [7]).

Higher-Order Mass

The accuracy of consistent mass can often be achieved by a much simpler lumped mass. A question naturally arises: Is consistent mass the best nondiagonal element mass? The answer appears to be: not always. There are examples of element mass matrices that exhibit superior accuracy to consistent mass. However, no general theory of obtaining higher-order accurate mass matrices exists yet.

Example 1

The first and simplest example of a higher-order accurate mass matrix was presented by Goudreau [8]. The matrix is to be viewed as an alternative for the two-node rod element. Recall the consistent and lumped mass matrices for this element are (respectively)

$$m_{\text{consistent}}^e = \frac{\rho h^e}{6} \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix} \quad (7.3.49)$$

$$m_{\text{lumped}}^e = \frac{\rho h^e}{2} \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \quad (7.3.50)$$

Both of these mass matrices result in second-order accurate eigenvalues. The higher-order mass matrix is simply the average of (7.3.49) and (7.3.50):

$$m_{\text{higher-order}}^e = \frac{\rho h^e}{12} \begin{bmatrix} 5 & 1 \\ 1 & 5 \end{bmatrix} \quad (7.3.51)$$

For a uniform mesh, it can be analytically shown that (7.3.51) leads to fourth-order accurate frequencies [8]. Numerical results supporting this assertion are presented for a fixed-fixed rod in Fig. 7.3.9. The same order of convergence is exhibited for a non-uniform mesh in Fig. 7.3.10 and for fixed-free boundary conditions in Fig. 7.3.11. Frequency spectra are presented in Fig. 9.1.4.)

For purposes of generalization to other elements it may be observed that (7.3.51) may be produced by employing the quadrature rule $\xi_1 = -\xi_2 = \sqrt{3}/3$, $W_1 = W_2 = 1$, on the element mass integral [10]. Clearly this rule can be iterated for multidimensional applications.

Another way of deriving (7.3.51) is to take a linear combination of the consistent mass and stiffness:

$$\begin{aligned} m^e &= m_{\text{consistent}}^e + \left(\frac{1}{6} - r \right) \frac{\rho (h^e)^2}{E} k^e \\ &= \frac{\rho h^e}{6} \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix} + \rho h^e \left(\frac{1}{6} - r \right) \begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix} \end{aligned} \quad (7.3.52)$$

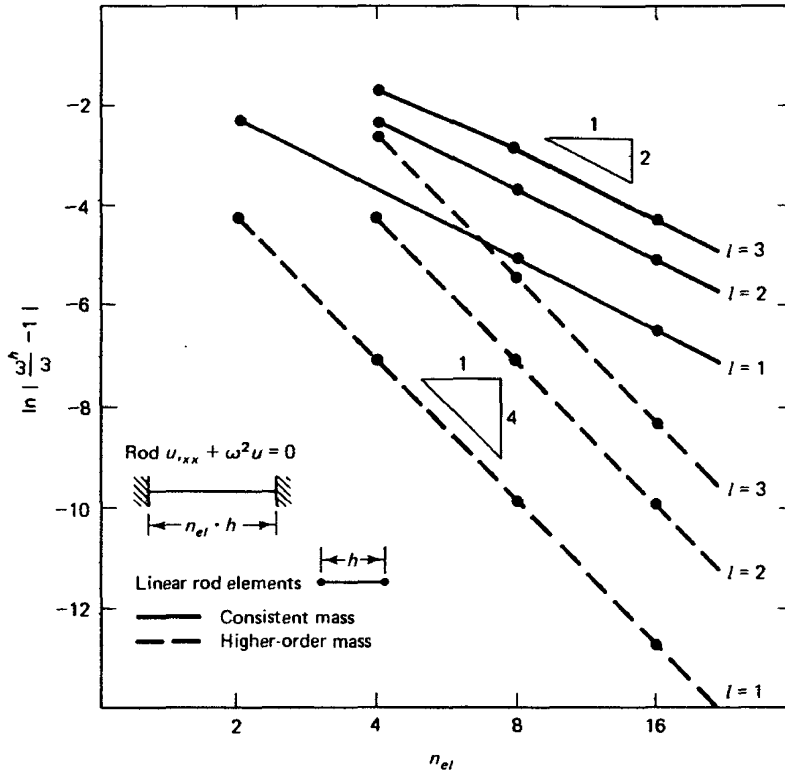


Figure 7.3.9 Convergence of the first three frequencies for linear rod elements [9].

Clearly, for $r = \frac{1}{12}$ we arrive at (7.3.51). The consistent and lumped mass matrices correspond to $r = \frac{1}{6}$ and $r = 0$, respectively.

Example 2

A higher-order mass matrix for the Hermite cubic beam element may be derived by taking a linear combination of the consistent mass and stiffness:

$$m_{\text{higher-order}}^e = m_{\text{consistent}}^e + \alpha \frac{\rho A (h^e)^4}{EI} k^e \quad (7.3.53)$$

Taylor and Iding [11] first experimented with matrices of this form, but did not obtain an optimal value of α . The value $\alpha = \frac{1}{720}$ was derived in [9] and shown to lead to sixth-order accurate frequencies for a uniform mesh. Numerical confirmation is presented in Fig. 7.3.12 for a simply supported beam. Frequency spectra are presented in Fig. 7.3.13.

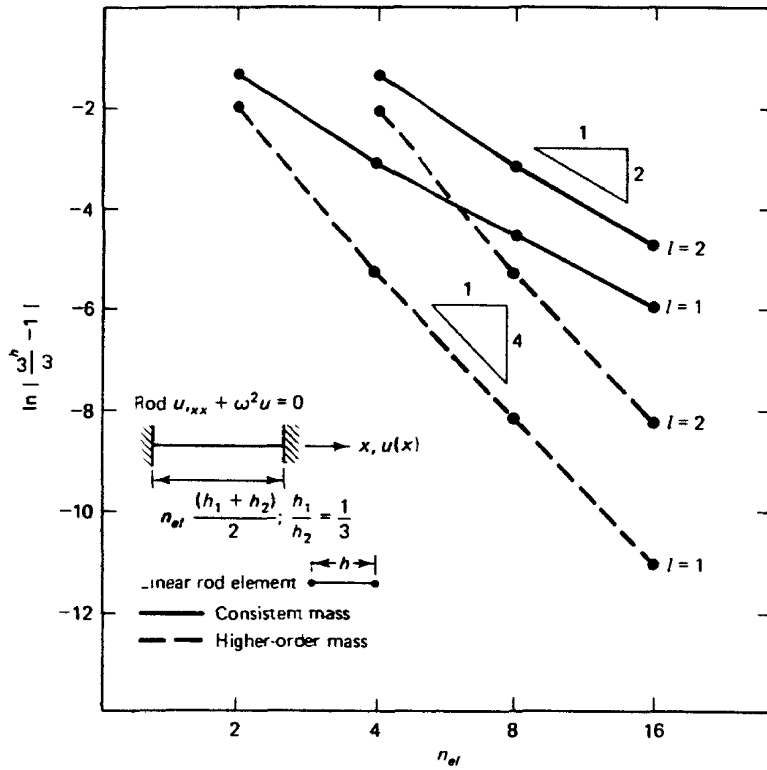


Figure 7.3.10 Convergence of the first two frequencies for linear rod elements with nonuniform mesh [9].

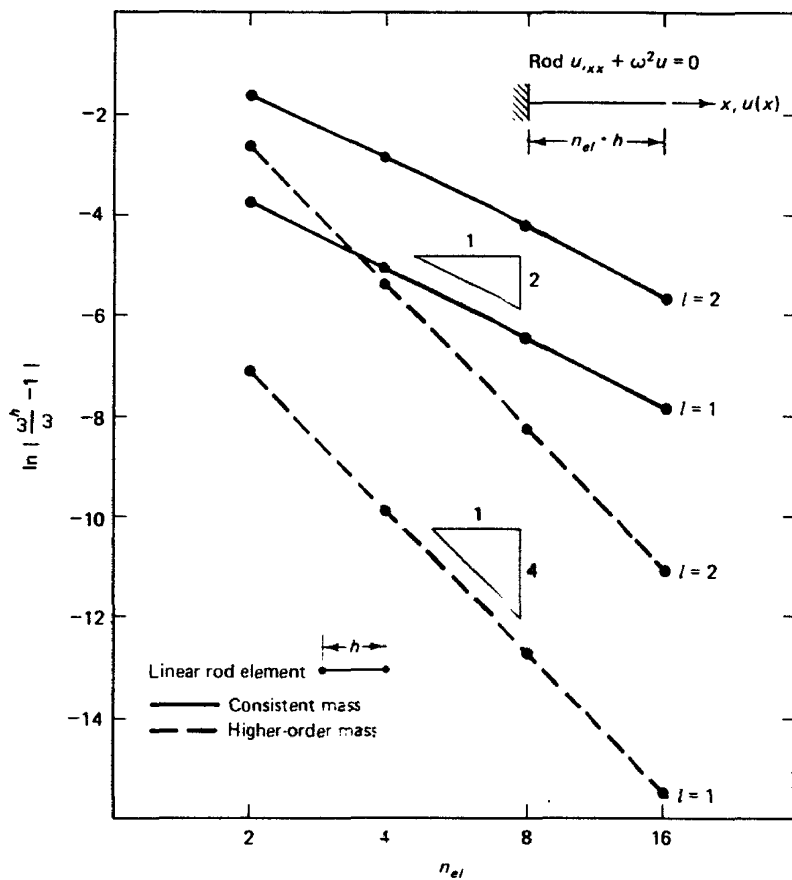


Figure 7.3.11 Convergence of the first two frequencies for linear rod elements with one end fixed and one end free [9].

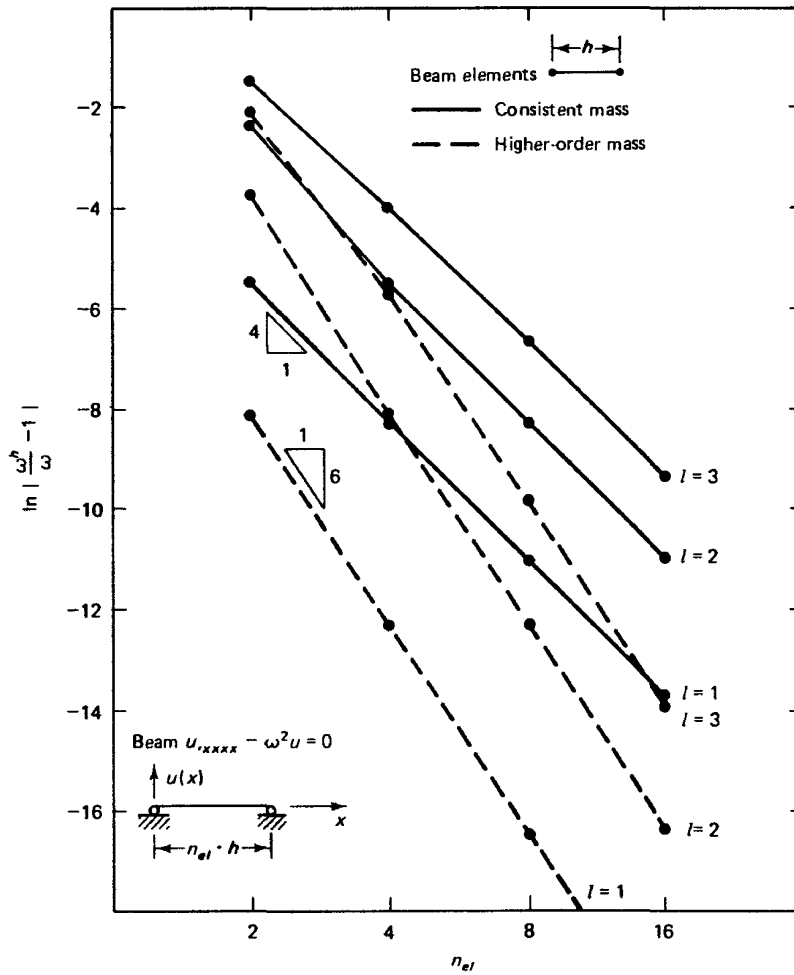


Figure 7.3.12 Convergence of the first three frequencies of a simply supported beam [9].

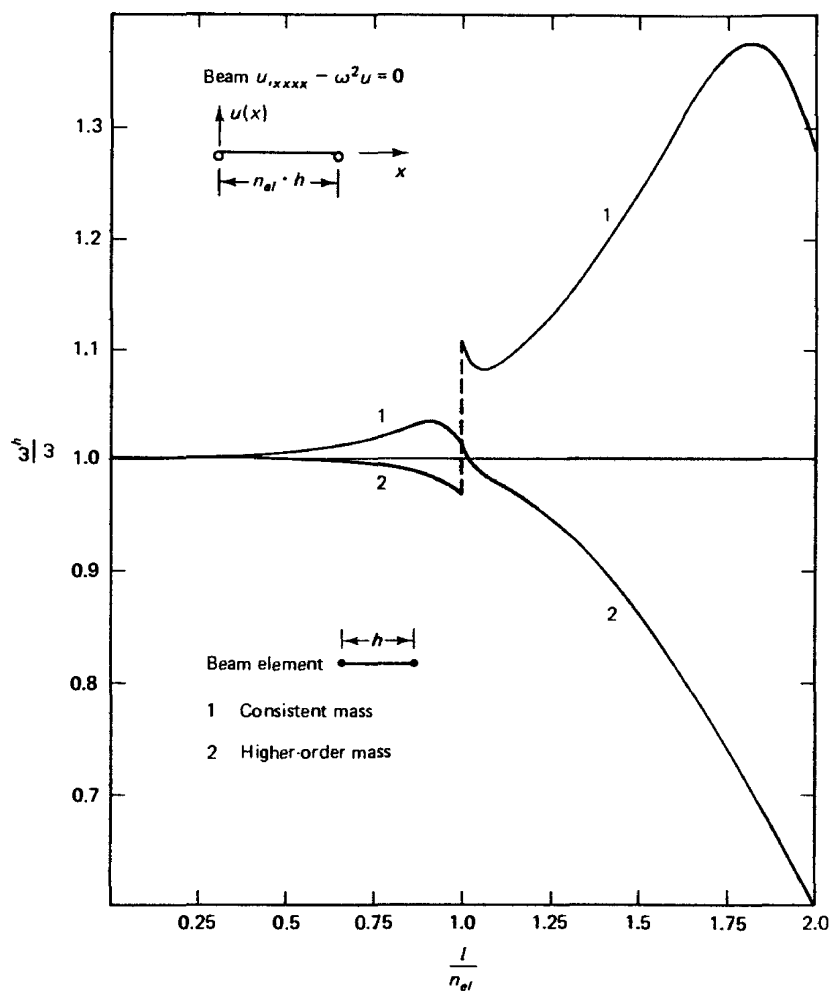


Figure 7.3.13 Frequency spectra for the cubic beam element [9].

7.3.3 Estimation of Eigenvalues

In time-dependent calculations it is often necessary to have a conservative estimate of the maximum eigenvalue (i.e., $\lambda_{n_{eq}}^h$). (We will return to this point in subsequent chapters.) It can be shown that [12,13]

$$\lambda_{n_{eq}}^h \leq \max_e (\lambda_{\max}^e) \quad (7.3.54)$$

where $\lambda_{\max}^e$ is the maximum eigenvalue of element e . The eigenvalues of element e satisfy²

$$(k^e - \lambda^e m^e) \psi^e = 0 \quad (7.3.55)$$

In the remainder of this subsection we will present an argument in support of (7.3.54).

Lemma

$$\lambda_{n_{eq}}^h = \max_{\substack{\phi \in \mathbb{R}^{n_{eq}} \\ \phi \neq 0}} \mathcal{R}(\phi) \quad (7.3.56)$$

where $\mathcal{R}(\phi) = \phi^T K \phi / \phi^T M \phi$ is the Rayleigh quotient, and the maximum occurs for $\phi = c \psi_{n_{eq}}$, where c is an arbitrary nonzero constant.

Proof. The eigenvectors $\psi_1, \psi_2, \dots, \psi_{n_{eq}}$ form a basis for $\mathbb{R}^{n_{eq}}$. Thus we can expand any vector, say ϕ , as

$$\phi = \sum_{l=1}^{n_{eq}} c_l \psi_l \quad (7.3.57)$$

and therefore

$$\phi^T M \phi = \sum_{l=1}^{n_{eq}} |c_l|^2 \quad (7.3.58)$$

$$\phi^T K \phi = \sum_{l=1}^{n_{eq}} \lambda_l^h |c_l|^2 \quad (7.3.59)$$

For $\phi \neq 0$,

$$\mathcal{R}(\phi) = \frac{\sum_{l=1}^{n_{eq}} \lambda_l^h |c_l|^2}{\sum_{l=1}^{n_{eq}} |c_l|^2} \leq \lambda_{n_{eq}}^h \quad (7.3.60)$$

by virtue of $\lambda_1^h \leq \lambda_2^h \leq \dots \leq \lambda_{n_{eq}}^h$. Because $\mathcal{R}(c \psi_{n_{eq}}) = \lambda_{n_{eq}}^h$, the proof of the lemma is complete. ■

²Solution of the element eigenvalue problem is actually rarely done. Whenever possible an exact formula is developed for $\lambda_{\max}^e$. Otherwise a conservative estimate is developed. This subject will be discussed further in Chapter 9.

Let K and M be global matrices formed by assembling the collections of matrices $\{k^e\}_1^{n_{el}}$ and $\{m^e\}_1^{n_{el}}$ according to some connectivity array (i.e., an "LM-array"; see Chapter 2). Consider the symmetric, block-diagonal matrices

$$\tilde{K} = \begin{bmatrix} k^1 & 0 & \cdots & 0 \\ 0 & k^2 & \ddots & \vdots \\ \vdots & \ddots & \ddots & 0 \\ 0 & \cdots & 0 & k^{n_{el}} \end{bmatrix} \quad (7.3.61)$$

$$\tilde{M} = \begin{bmatrix} m^1 & 0 & \cdots & 0 \\ 0 & m^2 & \ddots & \vdots \\ \vdots & \ddots & \ddots & 0 \\ 0 & \cdots & 0 & m^{n_{el}} \end{bmatrix} \quad (7.3.62)$$

The dimension of $\tilde{K}$ and $\tilde{M}$, say $\tilde{n}_{eq}$, clearly satisfies $\tilde{n}_{eq} \geq n_{eq}$.

The eigenvalue problems for the K , M , and $\tilde{K}$, $\tilde{M}$ systems are (respectively)

$$(K - \lambda^h M) \psi = 0 \quad (7.3.63)$$

and

$$(\tilde{K} - \tilde{\lambda}^h \tilde{M}) \tilde{\psi} = 0 \quad (7.3.64)$$

This lemma may also be applied to the eigenvalue problem (7.3.64). Thus

$$\tilde{\lambda}^h_{\tilde{n}_{eq}} = \max_{\substack{\tilde{\Phi} \in \mathbb{R}^{\tilde{n}_{eq}} \\ \tilde{\Phi} \neq 0}} \frac{\tilde{\Phi}^T \tilde{K} \tilde{\Phi}}{\tilde{\Phi}^T \tilde{M} \tilde{\Phi}} \quad (7.3.65)$$

Let $\mathbb{R}_c^{\tilde{n}_{eq}}$ denote the set of vectors $\tilde{\Phi} \in \mathbb{R}^{\tilde{n}_{eq}}$ such that components of $\tilde{\Phi}$ that correspond to the same degree of freedom in $\mathbb{R}^{n_{eq}}$ under the identification of the connectivity array used in assembling K and M are to be equal. With this definition, it is clear that

$$\lambda^h_{n_{eq}} = \max_{\substack{\tilde{\Phi} \in \mathbb{R}_c^{\tilde{n}_{eq}} \\ \tilde{\Phi} \neq 0}} \frac{\tilde{\Phi}^T \tilde{K} \tilde{\Phi}}{\tilde{\Phi}^T \tilde{M} \tilde{\Phi}} \leq \tilde{\lambda}^h_{\tilde{n}_{eq}} \quad (7.3.66)$$

Remarks

1. The equality in (7.3.66) may be obvious to some readers. If it is not, the following simple example should be helpful:

Let

$$\begin{aligned}
 k^e &= \begin{bmatrix} k_{11}^e & k_{12}^e \\ k_{21}^e & k_{22}^e \end{bmatrix} \\
 m^e &= \begin{bmatrix} m_{11}^e & m_{12}^e \\ m_{21}^e & m_{22}^e \end{bmatrix} \\
 e &= 1, 2 \\
 K &= \begin{bmatrix} k_{11}^1 & k_{12}^1 & 0 \\ k_{21}^1 & (k_{22}^1 + k_{11}^2) & k_{12}^2 \\ 0 & k_{21}^2 & k_{22}^2 \end{bmatrix} \\
 M &= \begin{bmatrix} m_{11}^1 & m_{12}^1 & 0 \\ m_{21}^1 & (m_{22}^1 + m_{11}^2) & m_{12}^2 \\ 0 & m_{21}^2 & m_{22}^2 \end{bmatrix} \\
 \underbrace{\widetilde{K}}_{4 \times 4} &= \begin{bmatrix} k^1 & 0 \\ 0 & k^2 \end{bmatrix} \\
 \underbrace{\widetilde{M}}_{4 \times 4} &= \begin{bmatrix} m^1 & 0 \\ 0 & m^2 \end{bmatrix} \\
 \Phi &= \begin{Bmatrix} \phi_1 \\ \phi_2 \\ \phi_3 \end{Bmatrix} \\
 \widetilde{\Phi} &= \begin{Bmatrix} \phi_1 \\ \phi_2 \\ \phi_2 \\ \phi_3 \end{Bmatrix} \in \mathbb{R}_c^4
 \end{aligned}$$

The equality in (7.3.66) is established for the example under consideration by the following calculations:

$$\begin{aligned}
 \widetilde{\Phi}^T \widetilde{K} \widetilde{\Phi} &= \begin{Bmatrix} \phi_1 \\ \phi_2 \end{Bmatrix}^T k^1 \begin{Bmatrix} \phi_1 \\ \phi_2 \end{Bmatrix} + \begin{Bmatrix} \phi_2 \\ \phi_3 \end{Bmatrix}^T k^2 \begin{Bmatrix} \phi_2 \\ \phi_3 \end{Bmatrix} \\
 &= \Phi^T K \Phi \\
 \widetilde{\Phi}^T \widetilde{M} \widetilde{\Phi} &= \begin{Bmatrix} \phi_1 \\ \phi_2 \end{Bmatrix}^T m^1 \begin{Bmatrix} \phi_1 \\ \phi_2 \end{Bmatrix} + \begin{Bmatrix} \phi_2 \\ \phi_3 \end{Bmatrix}^T m^2 \begin{Bmatrix} \phi_2 \\ \phi_3 \end{Bmatrix} \\
 &= \Phi^T M \Phi
 \end{aligned}$$

2. The inequality in (7.3.66) follows directly from (7.3.65).

In terms of the element matrices, the eigenvalue problem (7.3.64) can be written as

$$\begin{bmatrix} k^1 - \tilde{\lambda}^h m^1 & 0 & \dots & 0 \\ 0 & k^2 - \tilde{\lambda}^h m^2 & & \vdots \\ \vdots & & \ddots & 0 \\ 0 & \dots & 0 & k^{n_{el}} - \tilde{\lambda}^h m^{n_{el}} \end{bmatrix} \begin{Bmatrix} \tilde{\Psi}^1 \\ \tilde{\Psi}^2 \\ \vdots \\ \tilde{\Psi}^{n_{el}} \end{Bmatrix} = \begin{Bmatrix} 0 \\ 0 \\ \vdots \\ 0 \end{Bmatrix} \quad (7.3.67)$$

From this expression it can be seen that the eigenvalues of (7.3.64) are the same as the totality of eigenvalues of the uncoupled element problems

$$(k^e - \lambda^e m^e) \psi^e = 0, \quad e = 1, \dots, n_{el} \quad (7.3.68)$$

Therefore

$$\tilde{\lambda}_{\tilde{n}_{eq}}^h = \max_e (\lambda_{\max}^e) \quad (7.3.69)$$

where

$$\lambda_{\max}^e = \max_{\substack{\psi^e \in \mathbb{R}^{n_{ee}} \\ \psi^e \neq 0}} \frac{\psi^{eT} k^e \psi^e}{\psi^{eT} m^e \psi^e}$$

Combining (7.3.66) with (7.3.69), we have that

$$\lambda_{n_{eq}}^h \leq \max_e (\lambda_{\max}^e)$$

which completes the proof. ■